Predictive Coordination of Passive and Active Thermal Flexibility in a Multi-Unit Residential Building to Relieve a Distribution Transformer during Winter Peaks

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***Abstract*—The simultaneous electrification of heating and mobility makes the heat-pump (HP) peak and uncoordinated electric-vehicle (EV) charging coincide on winter evenings at the distribution transformers that supply multi-unit residential buildings. As an alternative to the prohibitive cost of reinforcing distribution-grid infrastructure, this paper proposes a two-layer hierarchical predictive coordination that jointly mobilizes the thermal inertia of the envelope, charged by the HPs (COP $\approx$ 3) before the peak, and refractory-brick electric thermal storage (ETS) units, which hold a lowered setpoint within the tolerated comfort band during the peak. To create neither an artificial peak nor a rebound, opportunistic ETS recharging is confined under a dynamic power envelope bounded by the off-peak maximum of the reference, and allocated in proportion to the dynamic storage deficit. Each dwelling agent solves a convex optimization problem based on a 4R2C multizone model coupled with a 2R2C ETS model. The physical thermal parameters of the model are calibrated recursively by an extended Kalman filter (EKF), fed with data from a digital twin of the building modeled in BEopt. The distributed hierarchical coordination strategy relies on two layers, namely an aggregator and agents (HEMS). The aggregator coordinates flexibility through a sequential automaton whose phases follow a signal based on a scarcity index that combines the transformer loading and its normalized aging rate (IEEE Std C57.91-2025). The case study addresses the digital twin of a 16-unit building supplied by a 50 kVA transformer over 72 hours. On a -17 °C day, the strategy reduces the peak by 19.3 %, lowers the maximum hot-spot temperature from 86.4 to 62.4 °C and removes 99 % of the discomfort, at the cost of a 4.5 % increase in daily energy.**

***Index Terms*—Distribution transformer, electric thermal storage, electric vehicles, model predictive control, multi-unit residential buildings, thermal flexibility, valley filling.**

NOMENCLATURE

Indices and sets

n, $j \in \mathcal{A}_n$, p, k	dwelling, adjacent dwelling, floor, time step
$\mathcal{C}$, σ	congestion set; off-congestion period

Transformer

S_n, PF, K_s	rated power (kVA); power factor; safety margin
K, R	load factor S/S_n; ratio of load to no-load losses
P_{lim}	active-power limit $(1-K_s)S_n\mathrm{PF}$ (kW)
θ_a, θ_{TO}, θ_{HS}	ambient, top-oil and hot-spot temperatures (°C)
$\Delta\theta_{TO,R}$, $\Delta\theta_{H,R}$	rated top-oil and hot-spot rises (K)
τ_{TO}, τ_w	oil and winding time constants
n, m	oil and winding exponents (ONAN)
F_{AA}, %LoL	aging acceleration factor; loss of life (%)

Dwelling and electric thermal storage (ETS)

T_i, T_e, T_{ext}	indoor-air, envelope and outdoor temperatures (°C)
$T_{h,p}$, α	corridor temperature of floor p; outdoor weight
T_c, T_s	ETS brick-core and casing temperatures
C_i, C_e, C_c, C_s	heat capacities: air, envelope, core, ETS casing (J/K)
R_i, R_o	air–envelope and envelope–outdoor resistances (K/W)
R_{jn}, R_h	resistances to the adjacent dwelling and to the corridor
R_c, R_s, R_{conv}^{min}	core–casing, casing–air and minimum forced-convection resistances
P_h, COP	HP electric power; coefficient of performance
P_{ETS}, Q_{ETS}	ETS charging power; heat released by the ETS (W)
Q_{int}, Q_0	occupancy gains; identified unmeasured internal gain
A_s, I_{gh}	equivalent solar aperture (m²); global horizontal irradiance (W/m²)
SOC, T_c^{min}, T_c^{max}	ETS state of charge; core temperature bounds
β, ξ, ψ	log-parameters; EKF augmented state; noise standard deviations

Coordination and control

P_B, P_{nf}, P_{coord}	reference load S0; non-flexible load (base + EV); coordinated load S1
E, L_σ, W_σ	dynamic envelope; valley-filling level; required recharge energy
M, c_n	building recharge margin; power cap of agent n
γ_n, γ_n^r, G_n	fixed share; dynamic recharge share; conductance to unheated zones
λ, λ_{pl}	scarcity index; floor cost
T^{ref}, T^{set}, T^{min}	reference and effective setpoints; comfort floor
T_j, ΔT_n	daytime setpoint; night setback
Δ_{pre}, Δ_{eff}, δ	preheating offset; peak setback; comfort tolerance
Δt, H	sampling period; prediction horizon (steps)

I. INTRODUCTION

The simultaneous electrification of heating and transport increases and synchronizes residential demand. In low-voltage networks, the joint deployment of HPs and EV chargers overloads distribution transformers [1]. Uncoordinated charging concentrates in the early evening [2]. Moreover, transformers, which are the bottlenecks of the network, undergo several overload-related stresses, the main consequence of which is aging, governed by the hot-spot temperature through an Arrhenius law [3], [4]. Coordinated charging can mitigate

this acceleration [5], and demand response combined with dynamic thermal rating increases the transformer reserve [6]. A poorly designed coordination may, however, **worsen** congestion: price-based scheduling of HPs and EVs synchronizes flexible loads to the point of exceeding the uncoordinated peak [7]. Relieving the transformer without reinforcement therefore calls on the building itself.

The thermal flexibility of buildings offers two sources:

- **Passive.** The envelope mass shifts heating [8]; in Québec, an MPC that preheats dwellings reduces the peak by about 30 % compared with deadband control [9], and near-optimal setpoint control cuts the peak of electrically heated schools by 40–65 % [10].
- **Active.** Refractory-brick ETS units charge a massive core off-peak and release heat through controlled convection [11], [12]. Their combination with building mass has been studied for single-family houses [13].

Mobilizing these sources at building scale then requires coordinating them. **Distributed coordination** of HEMS via ADMM enforces a common power limit [14]. Multizone MPC has been applied to dwellings and small buildings [15], [16]. For deferrable loads, valley filling is the reference for decentralized EV charging [17]. Table I positions these works against five criteria.

Table I reveals four gaps. **G1.** MUBs are usually modeled as a single zone, which neglects heat exchange between adjacent dwellings and with the corridor [18]. Inter-unit couplings are rarely identified in a verifiable way. **G2.** Passive (envelope, COP $\approx$ 3) and active (ETS, COP = 1) flexibility are rarely *ordered* according to their efficiency at multi-dwelling scale. **G3.** Thermal-storage recharging is left to the local optimizer. Existing architectures guarantee an instantaneous limit, but not the absence of a new peak or rebound over the whole trajectory [7]. **G4.** The transformer constraint is rarely linked to its thermal state.

To address these gaps, the contributions of this paper are:

1. a coupled 4R2C multizone dwelling model with a 2R2C ETS model, identified in three steps (augmented-state EKF, output error, maximum likelihood) with an identifiability condition for inter-unit couplings (G1);
2. a dynamic setpoint that **orders** the sources by efficiency (night setback, HP preheating of the envelope, then ETS-held peak setback) with *adaptive* HP curtailment (G2);
3. an **opportunistic recharge envelope** (bounded valley filling, dynamic deficit-based allocation) that provably creates neither an artificial peak nor a rebound (Proposition 1) (G3);
4. a 72 h evaluation for a 50 kVA transformer down to -17 °C that reveals the **physical ceiling** of thermal peak shaving and links coordination to the hot-spot temperature and loss of life (G4, partially).

Section II presents the transformer, dwelling and ETS models and their identification, Section III the coordination, Section IV the case study, Section V the results; Section VI concludes.

II. Modeling

The proposed coordination relies on three building blocks: the transformer model, which defines the constraint (II-A); the dwelling and ETS model, which describes the available flexibility (II-B); and the identification procedure that sets its parameters (II-C).

A. Transformer Thermal Model

We use the method of §7.3.8, Eq. (48), of IEEE Std C57.91-2025 [3], consistent with IEC 60076-7 [4]. With $K = S/S_n$ the load factor, and for K constant over a step Δt, the discrete solution reads:

$$\begin{aligned}\theta_{TO}(k{+}1) &= \theta_{TO}^{ss}(k) + \big(\theta_{TO}(k) - \theta_{TO}^{ss}(k)\big)e^{-\Delta t/\tau_{TO}},\\ \theta_{TO}^{ss} &= \theta_a + \Delta\theta_{TO,R}\left[\tfrac{K^2R+1}{R+1}\right]^n\end{aligned} \tag{1}$$

$$\begin{aligned}\Delta\theta_H(k{+}1) &= \Delta\theta_{H,R}K^{2m} + \big(\Delta\theta_H(k) - \Delta\theta_{H,R}K^{2m}\big)e^{-\Delta t/\tau_w},\\ \theta_{HS} &= \theta_{TO} + \Delta\theta_H\end{aligned} \tag{2}$$

with $n = m = 0.8$ for ONAN cooling (Table 6 of the standard). This exact form is unconditionally stable and removes the criterion $\Delta t \le \tau_w/2$ (§7.3.9) of explicit schemes. For thermally upgraded paper [3]:

$$\begin{aligned}F_{AA} &= \exp\Big(\frac{15\,000}{383.15} - \frac{15\,000}{\theta_{HS}+273.15}\Big),\\ \%LoL &= \frac{100}{180\,000}\sum_k F_{AA}(k)\Delta t\end{aligned} \tag{3}$$

The normal-life limits are $\theta_{HS} \le 120$ °C and $\theta_{TO} \le 105$ °C. This model provides the grid-side constraint; the building-side flexibility is described next.

B. Multizone Dwelling and ETS Model

Each dwelling n is described by its indoor air ($T_{i,n}$) and envelope ($T_{e,n}$). It is coupled to the adjacent dwelling $j \in \mathcal{A}_n$, to the corridor $T_{h,p}$ and to its ETS (core $T_{c,n}$, insulated casing $T_{s,n}$), as shown in Fig. 1 [19], [20]:

$$C_{e,n}\dot{T}_{e,n} = \frac{T_{ext} - T_{e,n}}{R_{o,n}} + \frac{T_{i,n} - T_{e,n}}{R_{i,n}} \tag{4}$$

$$\begin{aligned}C_{i,n}\dot{T}_{i,n} = {}& \frac{T_{e,n} - T_{i,n}}{R_{i,n}} + \sum_{j\in\mathcal{A}_n}\frac{T_{i,j} - T_{i,n}}{R_{jn}} + \frac{T_{h,p} - T_{i,n}}{R_{h,n}}\\ &+ \frac{T_{s,n} - T_{i,n}}{R_{s,n}} + \mathrm{COP}\,P_{h,n} + Q_{ETS,n} + Q_{int,n}\end{aligned} \tag{5}$$

$$\begin{aligned}C_{c,n}\dot{T}_{c,n} &= P_{ETS,n} - Q_{ETS,n} - \frac{T_{c,n} - T_{s,n}}{R_{c,n}},\\ C_{s,n}\dot{T}_{s,n} &= \frac{T_{c,n} - T_{s,n}}{R_{c,n}} - \frac{T_{s,n} - T_{i,n}}{R_{s,n}}\end{aligned} \tag{6}$$

TABLE I
POSITIONING (● ADDRESSED; ◐ PARTIALLY; ○ NOT ADDRESSED; × NOT ADDRESSED, SINGLE DWELLING; – NOT APPLICABLE).

Work	Coupled multizone MUB	Passive + active, coordinated	Transformer constraint	No-new-peak guarantee (trajectory)	Coordination
[14]	×	○ (passive only)	fixed limit	◐ (instantaneous)	iterative ADMM
[13]	×	◐ (not coordinated)	○	○	distributed
[9], [10]	×	◐	○ (price)	○	MPC / rules
[15], [16]	●	◐	○	○	local / central MPC
[6]	–	○	thermal (DTR)	◐	centralized
[17]	–	–	○	● (EV)	decentralized
This work	●	●	**fixed limit + aging**	● **(Prop. 1)**	**hierarchical, non-iterative**

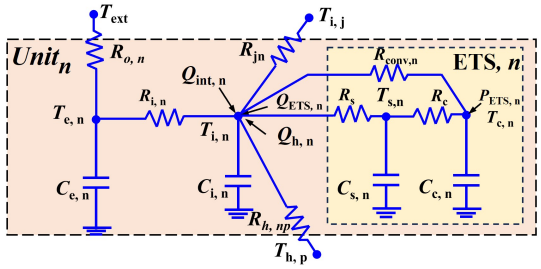


Fig. 1. Coupled RC model of a dwelling (4R2C) and its ETS (2R2C + controlled discharge). Dwelling nodes: air $T_{i,n}$ and envelope $T_{e,n}$; ETS nodes: core $T_{core} \equiv T_c$ and casing $T_{env} \equiv T_s$; controlled discharge through R_{conv}. Branches R_{jn} and $R_{h,np}$ connect the dwelling to its neighbor and to the corridor.

$$0 \le Q_{ETS,n} \le \min\left\{Q_{ETS}^{max}, \frac{T_{c,n} - T_{i,n}}{R_{conv}^{min}}\right\},$$
$$\mathrm{SOC}_n = \frac{T_{c,n} - T_c^{min}}{T_c^{max} - T_c^{min}} \tag{7}$$

where $T_{h,p} = \alpha T_{ext} + \frac{1-\alpha}{|\mathcal{N}_p|}\sum_{j\in\mathcal{N}_p} T_{i,j}$ for the corridor of floor p and $C_{c,n} = m_c c_p$ (brick mass and specific heat). The storage discharge $Q_{ETS,n}$ is a control input bounded by constraint (7), evaluated at the current state, which preserves convexity [11], [12]. With state $x_n = [T_c, T_s, T_i, T_e]^\top$, input $u_n = [P_h, Q_{ETS}, P_{ETS}]^\top$ and disturbances $d_n = [T_{ext}, Q_{int}, T_{h,p}, T_{i,j}]^\top$, the linear system (4)–(6) is discretized exactly with a zero-order hold (exponential of the augmented matrix):

$$x_n(k{+}1) = A_d x_n(k) + B_d u_n(k) + E_d d_n(k),$$
$$\begin{bmatrix} A_d & [B_d\ E_d] \\ 0 & I \end{bmatrix} = \exp\left(\begin{bmatrix} A & [B\ E] \\ 0 & 0 \end{bmatrix}\Delta t\right) \tag{8}$$

The capacitances and resistances of (4)–(5) cannot be measured directly and must be identified.

C. Identification of the Thermal Parameters

Data. The data come from a BEopt 2.8.0 digital twin (Standard mode, new construction, multi-family) [21] built on EnergyPlus, with the 2011–2025 TMYx weather file of Trois-Rivières [22]. The typical floor and its RC reduction are shown in Fig. 2. For each dwelling, two one-month experiments

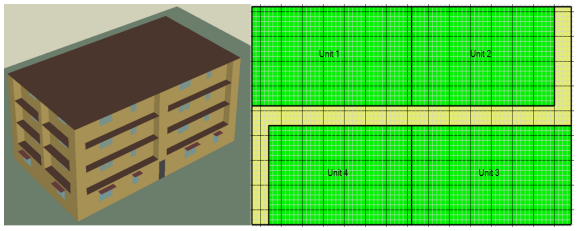


Fig. 2. 3D view of the BEopt digital twin model of the 16-unit MUB building distributed over four floors

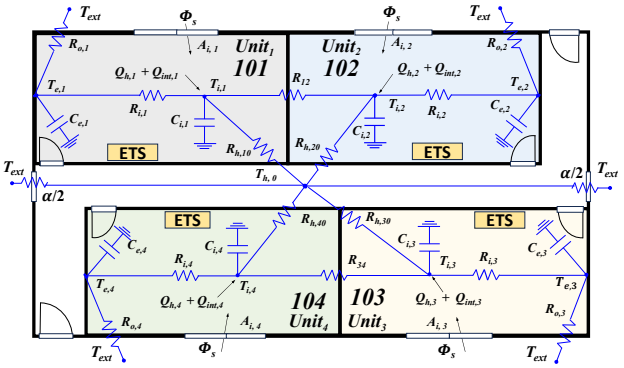


Fig. 3. 2D view and reduced 4R2C model of a unit of floor p of the MUB (units 101–104, central corridor): party-wall couplings R_{12}, R_{34}, corridor couplings $R_{h,n0}$ and solar apertures $A_{i,n}$.

are simulated (January, hourly): $\mathcal{E}_1$, variable PRBS setpoint heating (pulsed reheats to 25 °C followed by free decays), and $\mathcal{E}_2$, free-floating.

Identification model. Identification is decoupled: neighbor and corridor temperatures are measured inputs [20]. Submodel (4)–(5), without ETS, is augmented with two input coefficients: an equivalent solar aperture A_s applied to the global horizontal irradiance I_{gh}, and an unmeasured internal gain Q_0. These coefficients do not modify the dynamics A_d used by the controller. The free parameters $\beta = \log[R_i\ R_o\ C_i\ C_e\ A_s\ Q_0]$ are estimated in log scale, which guarantees positivity and balances magnitudes; the model is discretized exactly by (8).

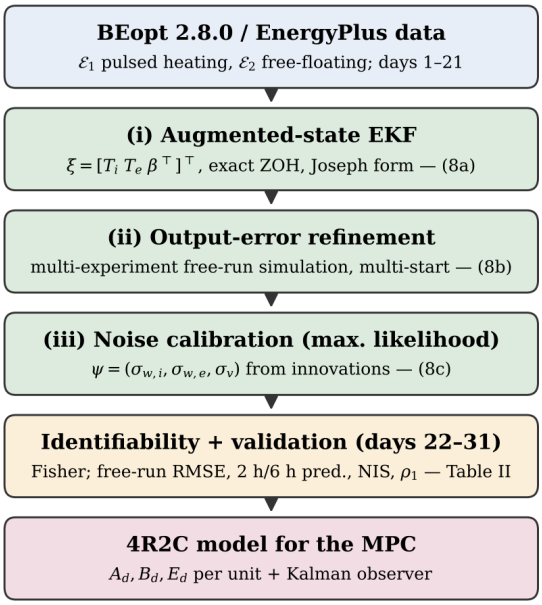


Fig. 4. Identification framework: augmented-state Extended Kalman Filter (EKF), output-error refinement, noise covariance tuning through maximum likelihood estimation, followed by an identifiability analysis and out-of-sample validation.

Three-step estimation. (i) An augmented-state EKF on $\xi_k = [T_i\ T_e\ \beta^\top]^\top$, where β follows a slow random walk, jointly estimates states and parameters from T_i only, the two experiments being chained [23], [24]; (ii) its overconfident estimate initializes an **output-error** (free-run) least-squares refinement, the relevant criterion for a prediction model [25]; (iii) the observer noise covariances are calibrated by maximum likelihood of the innovations [24].

Identifiability and protocol. The Fisher information matrix [26] yields two results. First, BEopt applies the same setpoint to all units (mean neighbor difference 0.48 °C); since the sensitivity of T_i to R_{jn} is driven by $T_{i,j} - T_{i,n}$, R_{jn} and R_h are not excited and are fixed at their geometric values. Second, on $\mathcal{E}_1$ alone, losses and gains are confounded ($r(R_o, Q_0) = -0.97$); multi-experiment calibration on days 1–21 of $\mathcal{E}_1$ and $\mathcal{E}_2$ brings $|r|$ down to 0.74–0.90 [19]. Validation on days 22–31 covers free-run simulation, 2 h (MPC horizon) and 6 h prediction with the Kalman observer, and observer consistency (NIS, lag-1 innovation autocorrelation ρ_1) (Table II).

On upper floors, all criteria are met. On the ground floor, the uninsulated slab couples the dwelling to the ground, which the 4R2C model does not represent: the free-run error is higher, but the prediction error at the MPC horizon remains below 0.8 °C because the observer corrects the state at every step. The validated model is the one used by each local agent of the coordination described next.

III. Hierarchical Coordination

The hierarchical coordination architecture is organized in two layers (Fig. 3): the first is the **grid aggregator**, located

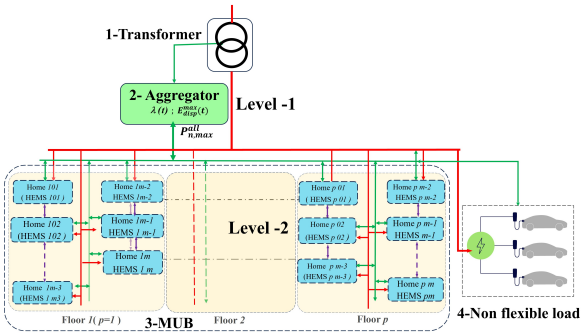


Fig. 5. Coordination architecture: two layers; the aggregator, at the transformer, broadcasts the index $\lambda(t)$ and the power cap $c_n \equiv P_{disp}^{max}(t)$ to the HEMS agents of each dwelling and floor; EV charging remains uncoordinated.

at the distribution transformer; the second consists of the **local HEMS-MPC agents**, installed in each apartment and described by Fig. 1. The aggregator computes *ex ante* the index λ, the automaton phases and the envelope $E(k)$, then allocates the margins; each HEMS-MPC agent optimizes its dwelling; a guard filter checks the aggregate. Subsections A–E follow the order of these computations.

A. Reference Trajectory and Scarcity Index

The aggregator first simulates the uncoordinated reference trajectory $P_B(k) = P_{nf}(k) + \sum_n P_{h,n}^B(k)$, where P_{nf} gathers the aggregated base load of the MUB and EV charging, and $P_{h,n}^B$ is the power of PID-controlled HPs tracking the same reference setpoint (Section III-B). This trajectory, assumed perfectly forecast, feeds (1)–(2) and the dimensionless index:

$$\lambda(k) = w_L \min\left\{1, e^{\kappa(u(k)-1)}\right\} + w_T \min\left\{1, \frac{F_{AA}(\theta_{HS}^B(k))}{F_{AA}(120\,^\circ\mathrm{C})}\right\},$$
$$u = P_B/P_{lim}, \quad P_{lim} = (1 - K_s) S_n \mathrm{PF} \tag{9}$$

with $w_L + w_T = 1$, which guarantees $\lambda \in [0, 1]$. The first term equals 1 at the safety limit and decays exponentially below it; the second is the **normalized aging rate**, equal to 1 when the predicted hot spot reaches 120 °C. It is broadcast over the horizon with a floor cost λ_{pl}, so that off-peak energy is not free and HP efficiency is valued against the ETS. The index tells *when* the grid is constrained; the dynamic setpoint defines *how* dwellings respond.

B. Dynamic Setpoint: Night Setback, Preheating and Peak Setback

The reference setpoint applies the night setback recommended in Canada—a 2–3 °C reduction during sleep [27]:

$$T^{ref}(k) = \begin{cases} T_j & h(k) \in [6{:}00,\ 22{:}01) \\ T_j - \Delta T_n,\ \Delta T_n \le 3\ ^\circ\mathrm{C} & \text{otherwise} \end{cases} \tag{10}$$

The automaton (Section III-C) then modulates the effective setpoint sent to the agents:

TABLE II
IDENTIFICATION ON THE TEST WINDOW (DAYS 22–31); PREDICTIONS, NIS AND ρ_1: AVERAGES OVER $\mathcal{E}_1$ AND $\mathcal{E}_2$.

Dwellings	**RMSE $\mathcal{E}_1$ (°C); R^2**	**RMSE $\mathcal{E}_2$ (°C); R^2**	**2 h pred. (°C)**	**6 h pred. (°C)**	**NIS**	ρ_1
Ground floor (101–104)	1.56; 0.74	1.66; 0.70	0.78	1.00	0.87	0.10
Upper floors (201–404)	1.07; 0.90	0.96; 0.94	0.78	0.95	0.95	0.13

$$T^{set}(k) = \begin{cases} T^{ref} + \Delta_{pre} & \text{preheating} \\ \min\{T^{ref},\, T_j - \Delta_{eff}\} & \text{congestion} \\ T^{ref} & \text{otherwise} \end{cases} \qquad (11)$$
$$T^{min}(k) = T^{set}(k) - \delta$$

This sequence orders the sources by efficiency. Before the peak, the HP (COP ≈ 3) stores heat in the envelope mass by raising the setpoint by Δ_{pre}. During the peak, the setpoint is lowered by Δ_{eff} within the tolerated band: the free decay of T_i releases the heat stored in the envelope, then the ETS (COP = 1) holds the lowered setpoint. The $\min\{\cdot\}$ prevents the peak setback from adding to the night setback when congestion extends beyond 22:01. Switching between setpoints is decided by the phase automaton.

C. Phase Automaton and Adaptive Curtailment

Phases are determined *ex ante* on the reference. The congestion set is $\mathcal{C} = \{k : P_B(k) \geq P_{lim}\}$, from which exceedances shorter than τ_{min} are removed (e.g., the morning reheat after the night setback); these are handled by the envelope (Section III-D) with $E(k) = \max\{P_{lim}, P_{nf}(k)\}$. Table III summarizes the bounds imposed at each horizon step j, so that each agent knows the upcoming curtailment periods in advance.

During congestion, the HP is not switched off but limited to its share of the residual transformer margin, allocated in proportion to each dwelling's conductance to unheated zones:

$$\gamma_n = \frac{G_n}{\sum_l G_l}, \qquad G_n = \frac{1}{R_{i,n} + R_{o,n}} + \frac{1}{R_{h,n}} \qquad (12)$$

When $P_{nf} > P_{lim}$, the margin is zero and the coordinated load equals P_{nf}: the peak reduction is then **bounded by the heating share of the reference peak**, $\Delta P_{max}/P_{B,max} \leq P_h^B(k^*)/P_B(k^*)$. Curtailment, however, creates a storage deficit in the ETS units, which must be refilled off-peak without creating a new peak: this is the role of the recharge envelope.

D. Opportunistic Recharge Envelope

ETS recharging must create neither an artificial peak nor a rebound, and should exploit the hours when the transformer is lightly loaded. The aggregator defines a dynamic envelope for the coordinated load:

$$E(k) = \begin{cases} P_B(k) & k \in \mathcal{C} \\ \max\left\{P_B(k),\ \min\{sP_{lim}, L_\sigma\}\right\} & k \in \sigma \end{cases} \qquad (13)$$

where σ is an off-congestion period (between two congestions) and L_σ its valley-filling level [17]: the smallest level that makes the required recharge energy W_σ available, bounded by the reference maximum over the period,

$$L_\sigma = \min\Big\{L \leq \min\left(sP_{lim}, \max_{k\in\sigma} P_B(k)\right) : \sum_{k\in\sigma}[L - P_B(k)]^+\Delta t \geq W_\sigma\Big\},$$
$$W_\sigma = \min\Big\{f_b\,\mathrm{COP}\sum_{k\in\mathcal{C}_\sigma^+}\sum_n P_{h,n}^B(k)\Delta t,\ W^{max}\Big\} \qquad (14)$$

$\mathcal{C}_\sigma^+$ is the next congestion, whose heat supplied by the reference HPs must be released by the ETS units; $f_b > 1$ covers standby losses; W^{max} is the total usable ETS capacity. L_σ is found by bisection, since $\sum_k [L - P_B]^+$ is monotone in L. At each step, the building recharge margin over the horizon is

$$M(k+j) = \Big[E(k+j) - P_{nf}(k+j) - \sum_n \hat{P}_{h,n}(k+j)\Big]^+ \qquad (15)$$

where $\hat{P}_{h,n}$ is the HP plan of agent n at the previous step, shifted by one step; heating thus has priority. The total margin is split into a heating part, at the fixed share γ_n, and a recharge part, at the **dynamic** share of the storage deficit:

$$c_n(k+j) = \gamma_n\big[E - P_{nf} - M\big]^+ + \gamma_n^r(k)\,M,$$
$$\gamma_n^r(k) = \frac{\gamma_n[\mathrm{SOC}^\star - \mathrm{SOC}_n(k)]^+}{\sum_l \gamma_l[\mathrm{SOC}^\star - \mathrm{SOC}_l(k)]^+} \qquad (16)$$

Proposition 1 (no artificial peak). If each agent satisfies $P_{h,n} + P_{ETS,n} \leq c_n$ off congestion and the bounds of Table III during congestion, then for all k: $P_{coord}(k) \leq E(k) \leq \max\{P_B(k), \min(sP_{lim}, \max_\sigma P_B)\}$. Hence the coordinated load never exceeds the reference during congestion and never exceeds, off congestion, the off-peak maximum of the reference.

Off congestion, $\sum_n \gamma_n = \sum_n \gamma_n^r = 1$ and (16) give $\sum_n c_n = [E - P_{nf} - M]^+ + M = E - P_{nf}$ whenever $E \geq P_{nf}$, hence $P_{coord} = P_{nf} + \sum_n (P_{h,n} + P_{ETS,n}) \leq E$. During congestion, $\sum_n P_{h,n} \leq [P_{lim} - P_{nf}]^+$ and $P_{ETS,n} = 0$, hence $P_{coord} \leq \max\{P_{lim}, P_{nf}\} \leq P_B$. The bound on E follows from (13)–(14).

TABLE III
PHASE AUTOMATON (EVALUATED AT EACH HORIZON STEP).

Phase	**Condition**	**Setpoint**	**Bounds imposed on agent n**
0 Normal	none of the others	T^{ref}	$P_h + P_{ETS} \le c_n$ (16); $P_{ETS} \le \gamma_n^r M$; $Q_{ETS} = 0$
1 Preheating	congestion within the next J_{pre} steps	$T^{ref} + \Delta_{pre}$	same as phase 0
2 Congestion	$k \in \mathcal{C}$	$\min\{T^{ref}, T_j - \Delta_{eff}\}$	$P_h \le \gamma_n [P_{lim} - P_{nf}]^+$; $P_{ETS} = 0$; Q_{ETS} by (7)
3 Recovery	J_{rep} steps after congestion	T^{ref}	$P_h \le \gamma_n [E - P_{nf}]^+$; $P_{ETS} = 0$; Q_{ETS} by (7)

A guard filter in the aggregator restores $P_{coord}(k) \le E(k)$, first by scaling down recharging, if an agent fails to converge.

Remark (strict variant). Imposing $P_{coord} \le P_B$ at every step ($L_\sigma = 0$) leaves almost no off-peak margin, since both trajectories track the same setpoint, whereas the heat released by the ETS must be bought back off-peak at COP = 1 (Section V-C). The automaton bounds and the cap c_n are the only information each local agent receives.

E. Local Model Predictive Control

With the condensed prediction matrices derived from (8), $T_{i,n} = c_T + G_T U_n$ and $T_{c,n}(k{+}H) = c_C + g_C^\top U_n$, where $U_n \in \mathbb{R}^{3H}$ is the input sequence, each agent solves:

$$\min_{U_n \ge 0} \sum_{j=0}^{H-1} \Big[\tilde{\lambda}_j (P_{h,n} + P_{ETS,n}) \Delta t + \mu_c \, e_j^2 + \mu_b \big[T_j^{min} - T_{i,n}\big]_+^2 + \mu_h \big[T_{i,n} - T_j^{max}\big]_+^2 + \mu_\Delta \|\Delta u_j\|^2 \Big] + \mu_T \big[\mathrm{SOC}^{ref} - \mathrm{SOC}_n(k{+}H)\big]_+^2 \tag{17}$$

subject to $u_j \le \bar{u}_j$ (Table III), $P_{h,n} + P_{ETS,n} \le c_n(k{+}j)$ and (7), with $e_j = T_{i,n} - T_j^{set}$ and $\tilde{\lambda} = \lambda + \lambda_{pl}$. Comfort is relaxed **asymmetrically** ($\mu_h \ll \mu_b$), since being warmer than the lowered setpoint is not discomfort. The terminal term gives stored energy a value (otherwise $P_{ETS} = 0$ is optimal): $\mathrm{SOC}^{ref} = 0$ when the horizon ends in congestion or recovery, $\mathrm{SOC}^\star$ otherwise, the envelope (13)–(16) deciding *when* to recharge. The penalties $[\cdot]_+^2$ are convex and continuously differentiable: (17) is a convex problem with 36 variables, solved by SQP (SLSQP in SciPy [28]) with an analytic gradient and an explicit envelope constraint. Only $\tilde{\lambda}$, c_n, the bounds and T^{set} are transmitted; no individual data are sent upstream, except the SOC used in (16). The next section applies this complete chain to a 16-unit building.

IV. CASE STUDY

The MUB has 16 dwellings on four levels (72 m² on the ground floor, 56 m² on upper floors). The coordination simulations use a **preliminary RC parameter set** with higher inertia than the set of Table II: envelope time constant $(R_i + R_o)C_e \approx 170$ h, versus 72 h on upper floors for the identified set; robustness to the identified set is discussed in Section V-E. The daytime setpoint is $T_j = 23$ °C with a night setback $\Delta T_n = 3$ °C (20 °C from 22:01 to 05:59) [27]. Internal gains follow a stochastic occupancy profile [29] (100 W per present occupant). The corridor of each floor is modeled algebraically with $\alpha = 0.8$.

The single-phase ONAN 50 kVA transformer (PF = 0.9, i.e., 45 kW) has losses $P_0 = 170$ W and $P_k = 390$ W, i.e., $R = 2.29$ (CSA C802). The limit is $P_{lim} = 0.95 \times 45 = 42.75$ kW and the ambient temperature θ_a is the outdoor temperature. Base-load, EV-charging (up to 36 kW in the early evening) and outdoor-temperature profiles cover 72 h, from mild to cold: day 1 from -3.0 to 3.7 °C, day 2 from -7.9 to 4.0 °C, and day 3, a cold snap, from -17.0 to -8.0 °C. The step is $\Delta t = 600$ s and the horizon $H = 12$ steps (2 h). The three days are analyzed separately; the ETS units start at $T_c = 500$ °C (SOC ≈ 78 %).

Two scenarios are compared on the same reference setpoint (10). **S0**: PID thermostats ($K_p = 1.9$, $K_i = 0.005$, $K_d = 0.02$, anti-windup), ETS inactive, uncoordinated EVs. **S1**: the proposed coordination. The strict variant of S1 ($L_\sigma = 0$) is also evaluated. Table IV lists the parameters.

Indicators. The peak window is set on S0. Rebound is the S1 energy above S0 during recovery and the following 2 h; valley-filling energy is the remaining off-peak excess. The daily energy of S1 is corrected for the change in ETS storage over the day, valued at COP = 1. Discomfort is the integral, summed over dwellings, of $[T^{min} - T_{i,n}]^+$. The flexibility contribution index (FCI) of a dwelling is its share of the λ-weighted net curtailment; fairness is measured by Jain's index [30] of the ratio FCI$/\gamma_n$.

V. RESULTS AND DISCUSSION

A. Peak Reduction and Physical Ceiling

Every evening, the reference peak occurs at 20:30, when EV charging (up to 36 kW) adds to the base load and the HPs (Fig. 4). Coordination reduces it from 66.2 to 56.4 kW (-14.8 %) on day 1, from 59.4 to 50.6 kW (-14.8 %) on day 2 and from 64.2 to 51.8 kW (-19.3 %) on day 3 (Table V). In all three cases, the reduction is **exactly equal** to the heating share of the reference peak, and the coordinated peak coincides with the maximum of P_{nf}. At peak time, the non-flexible load alone exceeds P_{lim}: the margin in (12) is zero, the HPs are fully curtailed and thermal flexibility is exhausted. The coordinated peak therefore remains above P_{lim} (load factor 1.15 on day 3) and the overload duration only drops from 4.7 to 3.7 h. This **physical ceiling** is structural: beyond it, only acting on EV

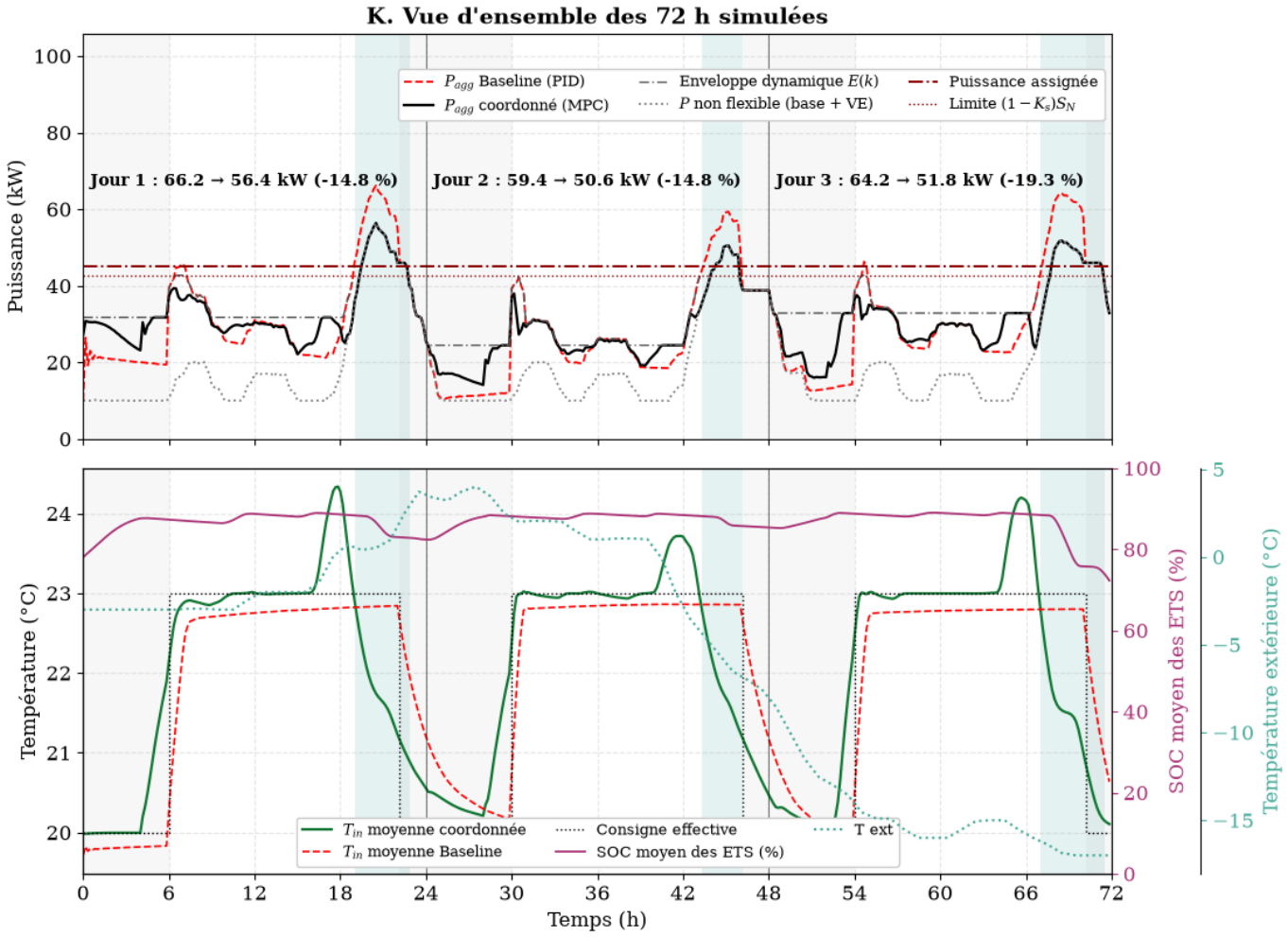


Fig. 6. Overview of the 72 simulated hours. Top: transformer load for S0 (PID, red dashed) and S1 (coordinated MPC, black), dynamic envelope $E(k)$, non-flexible load (base + EV, dotted), rated power and limit $(1-K_s)S_n$; green bands: congestion; gray bands: night setback. Bottom: mean indoor temperature for S0 and S1, effective setpoint, mean ETS SOC and outdoor temperature.

TABLE IV
MAIN PARAMETERS.

Item	Values
HP	$P_h^{max} = 1.8$ kW; COP = 3.0
ETS	$C_c = 105 \times 1155$ J/K; $C_s = 18$ kJ/K; $R_c = 12$ K/W; $R_s = 6$ K/W; $T_c \in [150, 600]$ °C; $P_{ETS}^{max} = Q_{ETS}^{max} = 2.5$ kW; SOC$^\star = 0.9$
Transformer	$S_n = 50$ kVA; PF = 0.9; $K_s = 5$ %; $\Delta\theta_{TO,R} = 55$ K; $\Delta\theta_{H,R} = 25$ K; $\tau_{TO} = 180$ min; $\tau_w = 5$ min; $n = m = 0.8$
Setpoint	$T_j = 23$ °C; $\Delta T_n = 3$ °C; $\Delta_{pre} = \Delta_{eff} = 1.5$ °C; $\delta = 0.5$ °C; $J_{pre} = 2$ h; $J_{rep} = 1$ h; $\tau_{min} = 1$ h
Envelope	$s = 0.90$; $f_b = 1.3$
Index λ	$w_L = 0.75$; $\kappa = 8$; $\lambda_{pl} = 0.30$; $w_T = 0.25$
MPC	$\mu_c = 2$; $\mu_b = 500$; $\mu_h = 2$; $\mu_\Delta = 0.05$; $\mu_T = 200$

charging or a dynamic thermal constraint can reduce the peak further.

B. Envelope-then-ETS Sequence

Fig. 5 details day 3. Preheating by 1.5 °C precedes congestion (Fig. 4, bottom, around 17:00–19:00); during congestion, the free decay of T_i toward the lowered setpoint first releases the envelope heat, then the ETS takes over (Fig. 5-C, discharge in magenta). The electric energy curtailed during the peak reaches 45.6 kWh, i.e., about 137 kWh of heat at COP 3; the ETS discharge supplies 37.1 kWh$_{th}$ of it, the rest coming from the envelope and the setpoint reduction. The corrected energy increase (+1.4 to +4.5 %) reflects the HP → ETS substitution (COP 3 → 1), moderated by the envelope support at COP 3. Comfort is **improved** (−82 to −99 % discomfort), mainly

TABLE V
SUMMARY OF INDICATORS PER DAY (S0 → S1).

Indicator	Day 1	Day 2	Day 3
Peak $\max P$ (kW)	66.2 → 56.4	59.4 → 50.6	64.2 → 51.8
Reduction; heating share (%)	14.8; 14.8	14.8; 14.8	19.3; 19.3
Curtailed energy; rebound (kWh)	37.6; 0.0	24.7; 0.0	45.6; 0.0
$\max(P_{coord} - E)$ (kW)	0.00	0.00	0.00
Off-peak maximum (kW)	41.8 → 39.9	42.6 → 38.9	41.8 → 40.3
$\max \theta_{HS}$ (°C)	101.5 → 83.3	87.9 → 71.5	86.4 → 62.4
Discomfort (°C·h)	14.94 → 2.74	10.83 → 0.34	12.22 → 0.11
ETS recharge (kWh_e); off-peak share (%)	37.3; 74	23.8; 83	16.7; 88
ETS discharge (kWh_{th}); min SOC (%)	11.9; 66	4.6; 66	37.1; 54
Corrected energy (kWh); change (%)	769.2 → 782.2; +1.7	644.4 → 653.5; +1.4	753.0 → 786.6; +4.5
Jain's index	0.980	0.991	0.997

because the MPC anticipates the 06:00 reheat that the PID performs as a delayed step (Fig. 4, bottom).

C. Opportunistic Recharging and Strict Variant

Off-peak, the coordinated load fills the valleys of the reference—mainly during the night setback, when reference HPs are almost idle—without ever exceeding the envelope: $\max(P_{coord} - E) = 0.00$ kW on all three days, which numerically confirms Proposition 1. Locally, P_{coord} may exceed P_B by 12.6 to 18.9 kW, but only within these valleys, and the off-peak maximum actually **decreases** (41.8 → 40.3 kW on day 3) because the MPC anticipates the 06:00 reheat that the PID concentrates into a step. No rebound is observed, and 74–88 % of ETS recharging occurs off-peak (Table V). The strict variant confirms the Remark of Section III-D: without valley filling, the off-peak margin is nearly zero, the ETS units receive only 1.0–3.8 kWh_e per day and become depleted (minimum SOC of 39 %, 17 % and then 0 %). The peak is the same as in S1, but discomfort reaches 80.5, 100.8 and 169.1 °C·h, i.e., 29 to more than 1,500 times that of S1: bounded valley filling is the condition for the strategy to be sustainable.

D. Transformer Thermal Signal

Coordination lowers the maximum θ_{HS} by 16.4–24.0 K depending on the day (Table V, Fig. 5-D). In winter, however, θ_{HS} stays far below 120 °C even in S0, the cold ambient cooling the unit (its maximum, 101.5 °C, occurs on the **mildest** day), and the loss of life stays below 5×10^{-4} %/day versus 0.013 %/day for normal aging. The aging term of (9) is thus inactive and the binding constraint is the **current**, which motivates a dynamic-thermal-rating envelope allowing temporary overloads in cold weather.

E. Fairness and Robustness

Ground-floor (slab-on-grade) and top-floor (roof) dwellings, which have the largest conductance G_n, contribute more (6.8–7.4 % each on day 3) than those on intermediate floors (4.6–5.9 %). The FCI closely follows the share γ_n: Jain's index is 0.980, 0.991 and 0.997 depending on the day.

Finally, with the identified set of Table II and a retuned PI reference, the coordinated peak is identical (set by P_{nf}) and the reduction still equals the heating share of the reference peak (10–16 % vs. 15–19 %); the lighter envelope relies more on the ETS (34–88 vs. 5–37 kWh_{th}/day, minimum SOC 37 % at −17 °C). The conclusions are thus robust to the parameter set.

VI. CONCLUSION

This paper proposed a hierarchical predictive coordination that successively mobilizes the envelope of a multi-unit residential building, preheated by HPs, and refractory-brick electric thermal storage units to relieve a 50 kVA distribution transformer during winter peaks. Three results emerge from the 72 h simulation of a 16-unit MUB. First, the peak is reduced by 14.8–19.3 %, exactly the heating share of the reference peak: when EV charging alone exceeds the transformer capacity, thermal flexibility reaches a physical ceiling. Second, the maximum hot-spot temperature drops by 16–24 K, but remains far below 120 °C in winter, so the active constraint is the current rather than aging. Third, these gains are obtained without artificial peak or rebound, with improved comfort (82–99 % less discomfort), high fairness (Jain's index ≥ 0.98) and a 1.4–4.5 % energy penalty due to the HP → ETS substitution. For a distribution utility, ETS recharging can thus be confined under the reference without creating an artificial peak, provided bounded valley filling is allowed; the strict variant, which forbids it, depletes the storage units.

These results rely on perfect forecasts, a preliminary parameter set and deterministic EV profiles. Future work will address a dynamic power envelope derived from the transformer thermal model, joint coordination of EV charging, stochastic control under forecast errors [31], a comparison of identification methods under parameter degradation, and hardware-in-the-loop validation.

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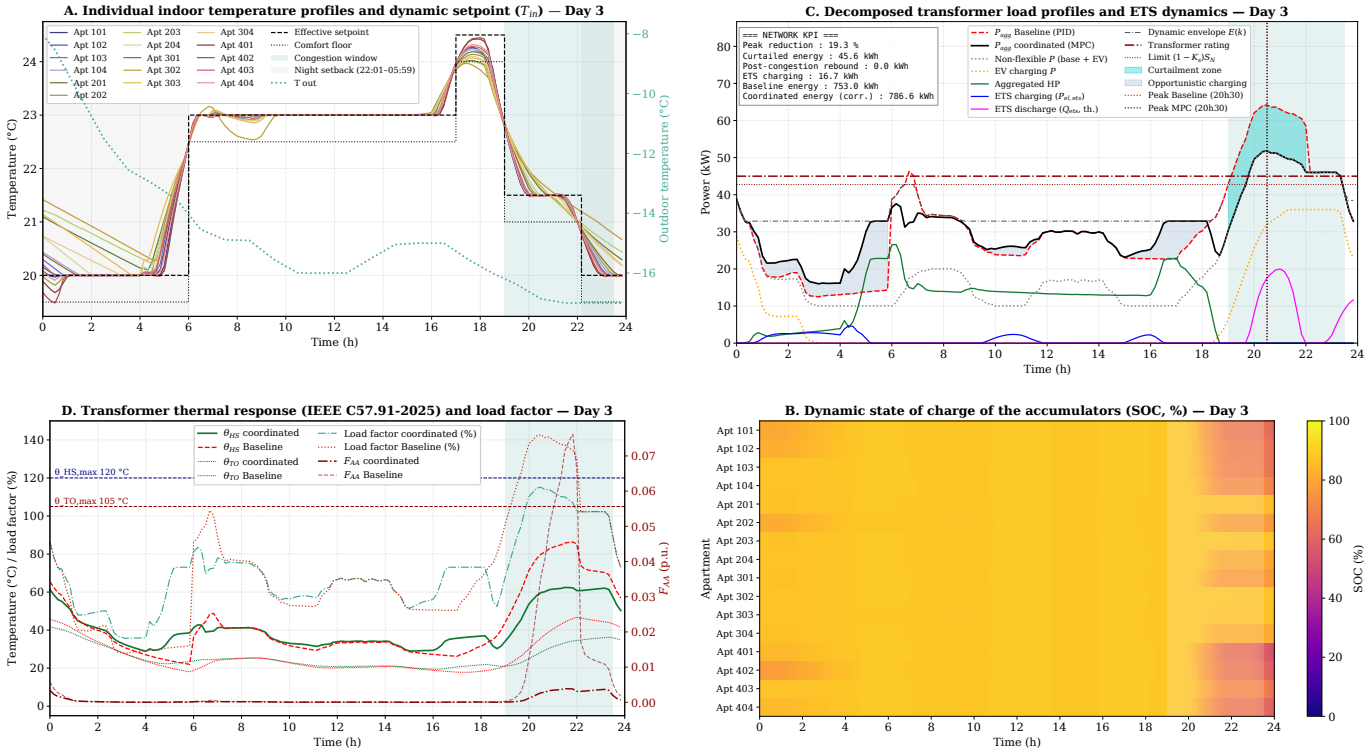


Fig. 7. Coordination mechanisms on day 3 (cold snap, −17 to −8 °C). (C) Transformer load decomposition: S0, S1, envelope $E(k)$, curtailment zone (cyan), opportunistic charging, ETS discharge and aggregated HPs. (D) Transformer thermal response (IEEE C57.91-2025): θ_{TO}, θ_{HS}, load factor and F_{AA} for S0 and S1. Green bands: congestion window.

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